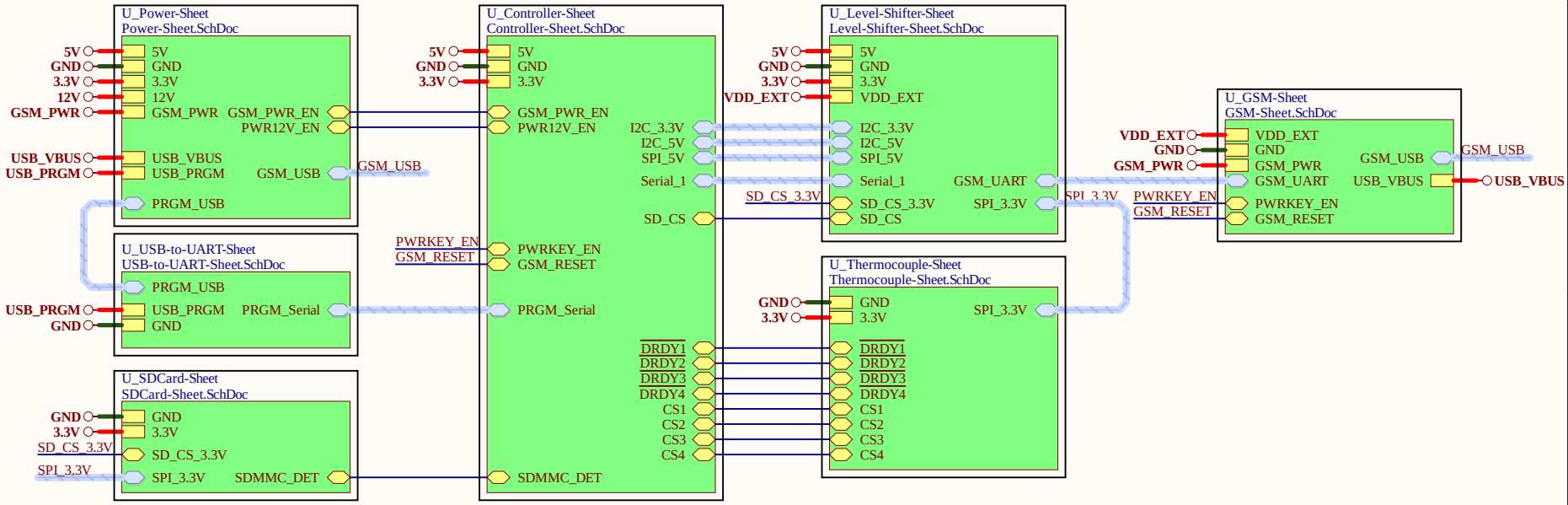
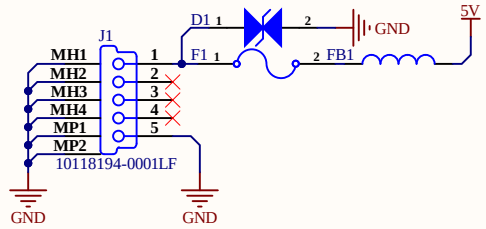


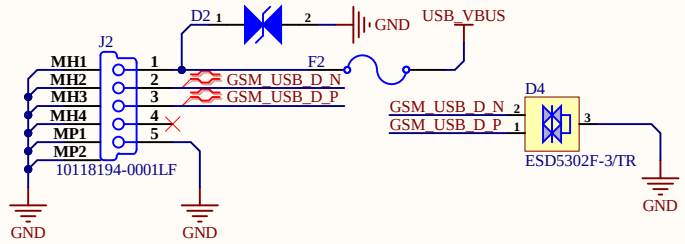
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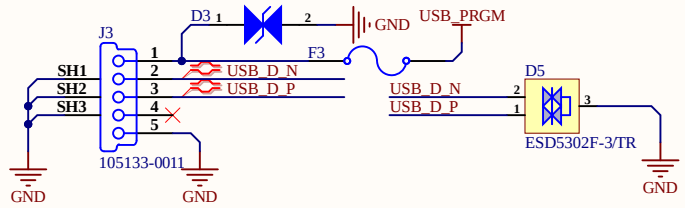
USB to Power Board



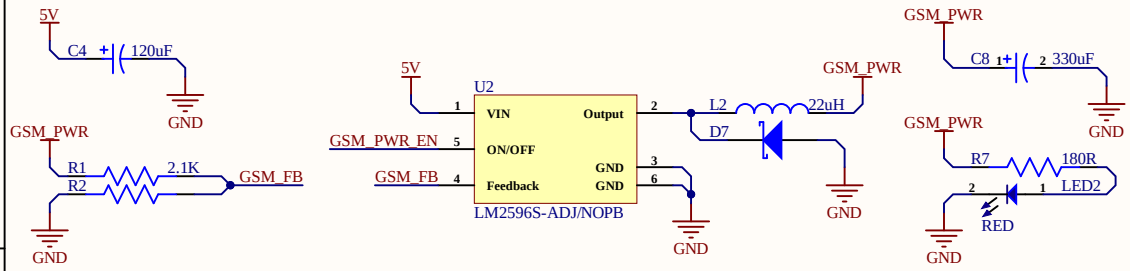
USB for GSM Updates



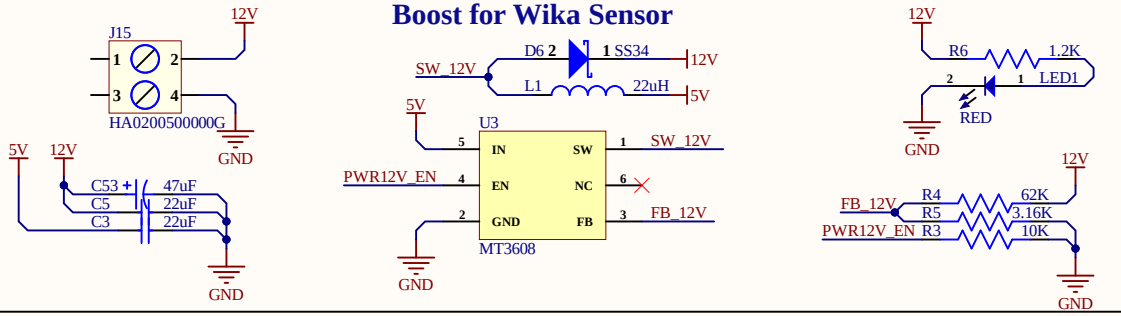
USB for Programming Mega2560



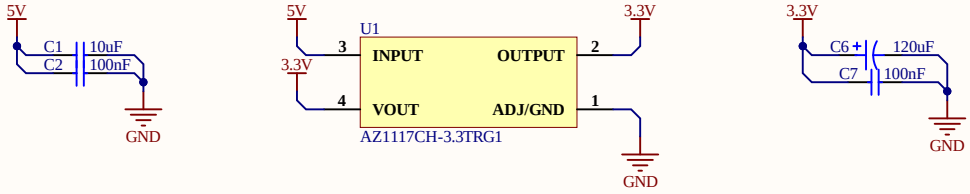
Buck for GSM Power



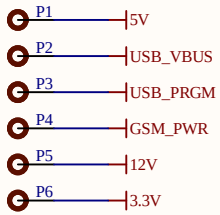
Boost for Wika Sensor

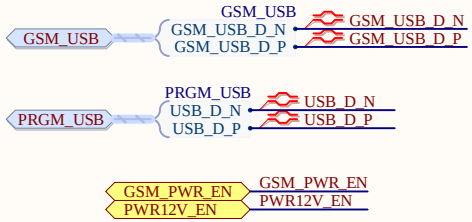


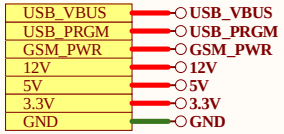
LDO 3.3V



Test Points



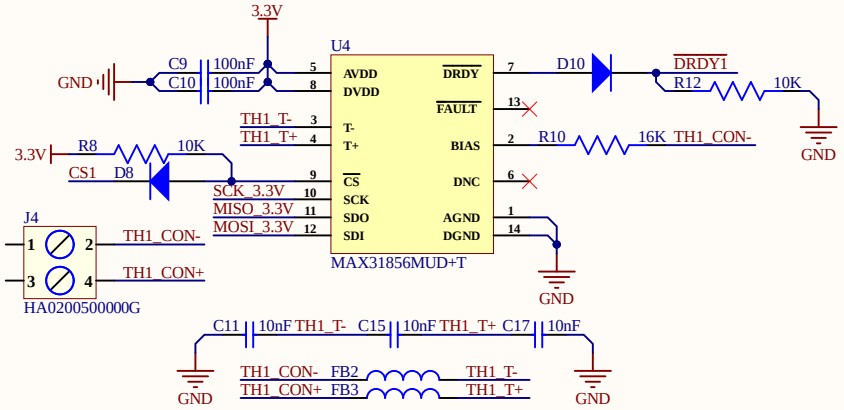




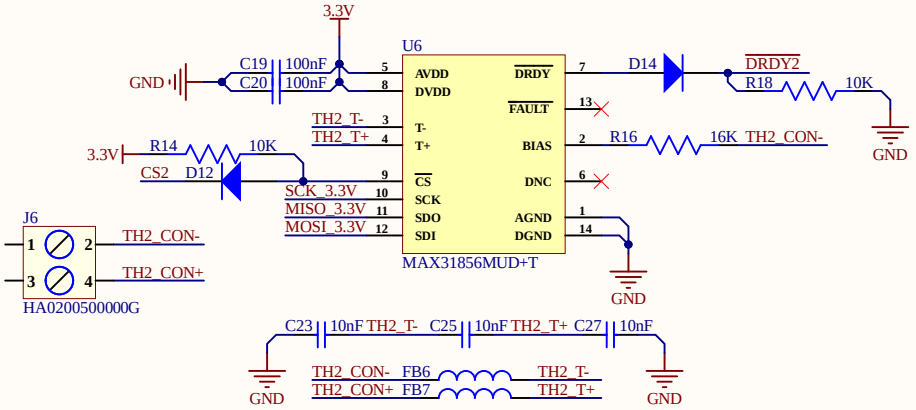
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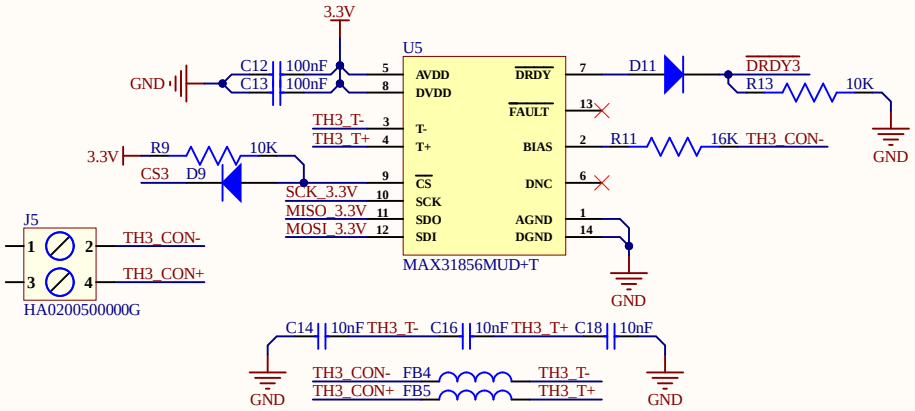
Thermocouple Input - 1



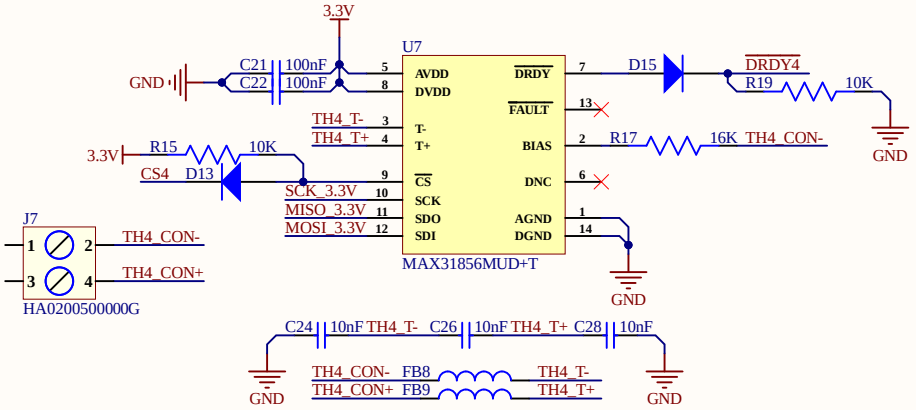
Thermocouple Input - 2



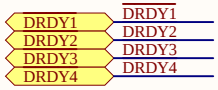
Thermocouple Input - 3

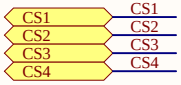


Thermocouple Input - 4







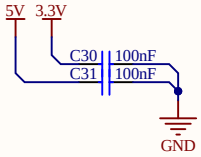


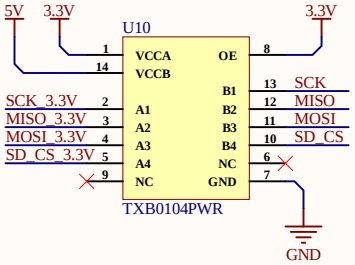


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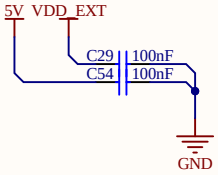
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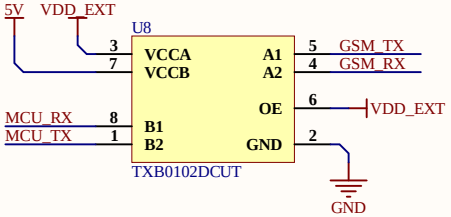
Level Shifter for SPI



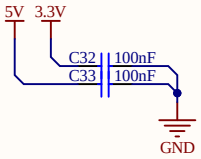


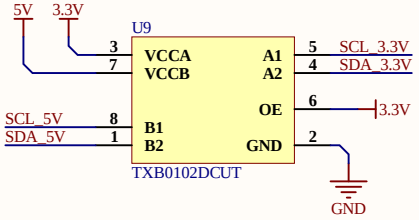
Level Shifter for Serial





Level Shifter for I2C

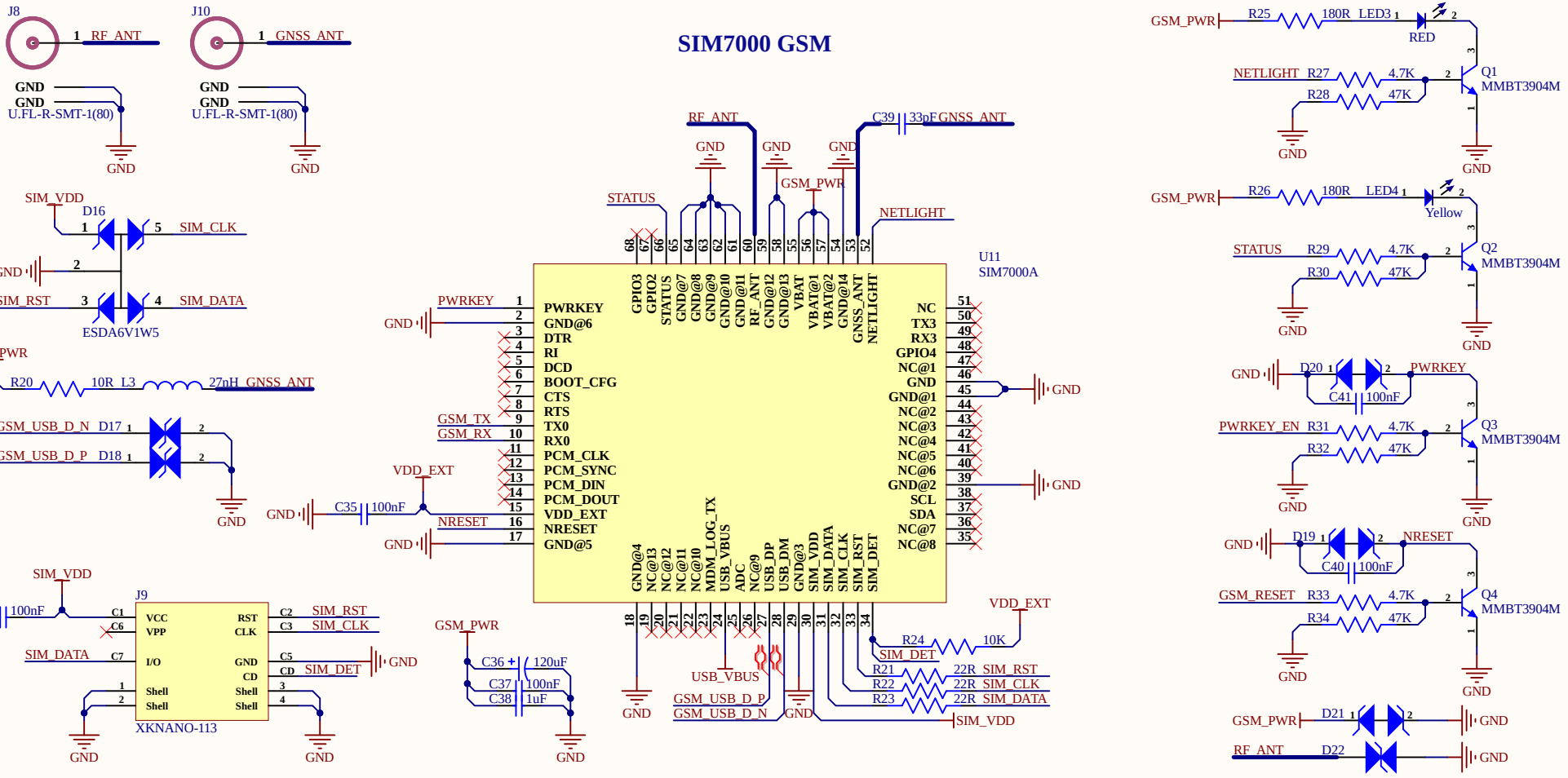


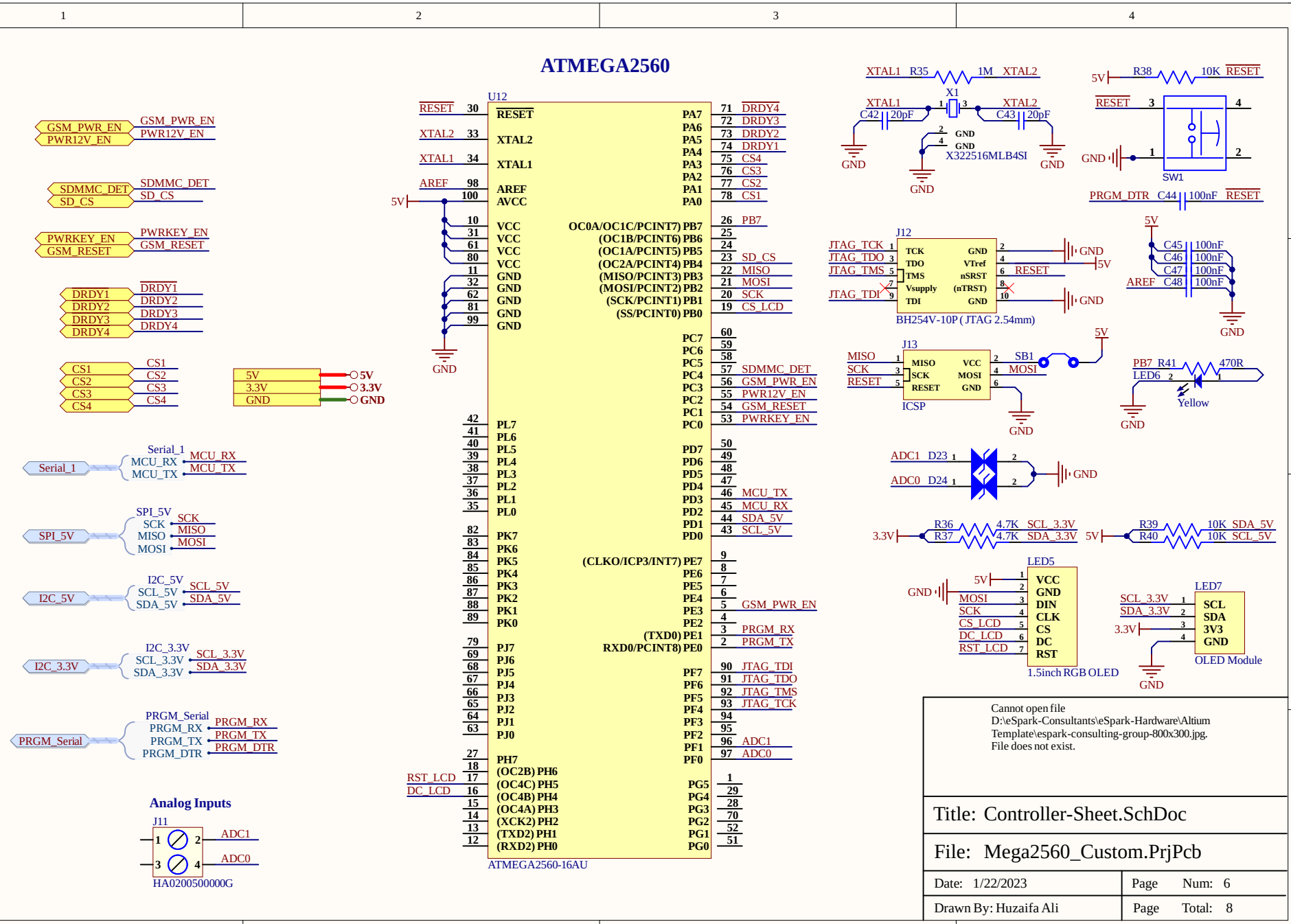




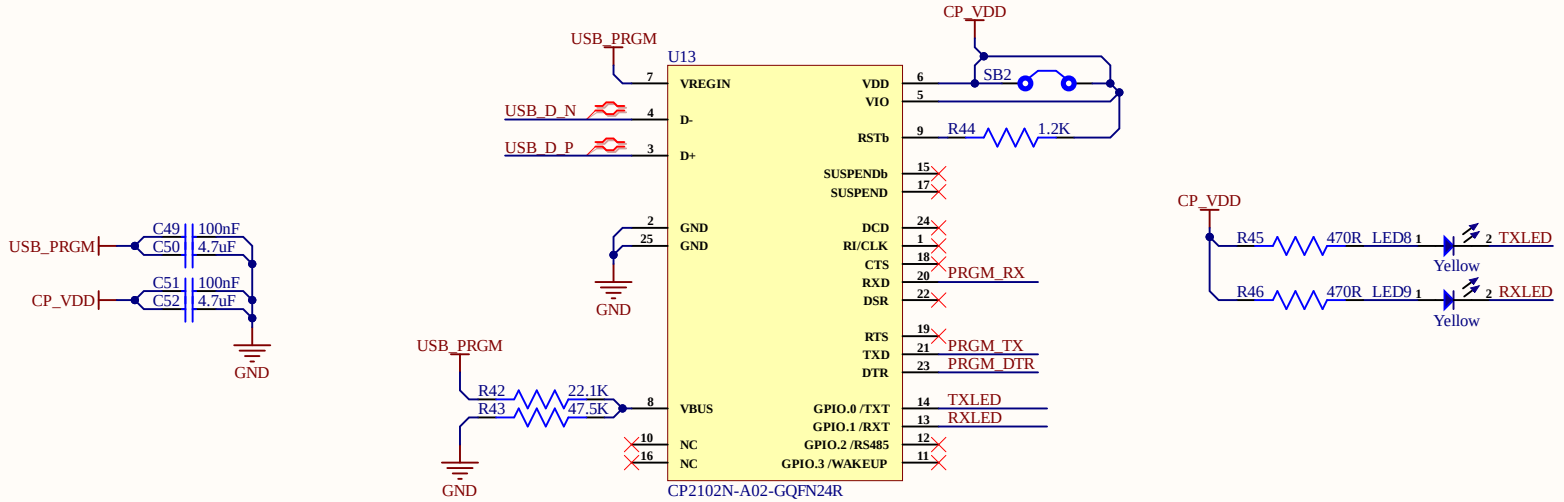
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SIM7000 GSM
J8
J10
RF_ANT
GNSS_ANT
GND
U.FL-R-SMT-1(80)
SIM_VDD
D16
SIM_CLK
SIM_RST
SIM_DATA
ESDA6V1W5
GSM_PWR
R20
10R
L3
27nH_GNSS_ANT
GSM_USB_D_N
D17
GSM_USB_D_P
D18
C35
100nF
VDD_EXT
NRESET
STATUS
NETLIGHT
PWRKEY
GSM_TX
GSM_RX
U11
SIM7000A
PWRKEY
GND@6
DTR
RI
DCD
BOOT_CFG
CTS
RTS
TX0
RX0
PCM_CLK
PCM_SYNC
PCM_DIN
PCM_DOUT
VDD_EXT
NRESET
GND@5
GPIO3
GPIO2
STATUS
GND@7
GND@8
GND@9
GND@10
GND@11
RF_ANT
GND@12
GND@13
VBAT
VBAT@1
VBAT@2
GND@14
GNSS_ANT
NETLIGHT
NC
TX3
RX3
GPIO4
NC@1
GND
GND@1
NC@2
NC@3
NC@4
NC@5
NC@6
GND@2
SCL
SDA
NC@7
NC@8
GND@4
NC@13
NC@12
NC@11
NC@10
MDM_LOG_TX
USB_VBUS
ADC
NC@9
USB_DP
USB_DM
GND@3
SIM_VDD
SIM_DATA
SIM_CLK
SIM_RST
SIM_DET
C34
100nF
J9
VCC
VPP
I/O
RST
CLK
GND
CD
Shell
XKNANO-113
C36
120uF
C37
100nF
C38
1uF
R24
10K
R21
22R
SIM_RST
R22
22R
SIM_CLK
R23
22R
SIM_DATA
USB_VBUS
GSM_USB_D_P
GSM_USB_D_N
R25
180R
LED3
RED
NETLIGHT
R27
4.7K
R28
47K
Q1
MMBT3904M
R26
180R
LED4
Yellow
STATUS
R29
4.7K
R30
47K
Q2
MMBT3904M
D20
C41
100nF
PWRKEY
PWRKEY_EN
R31
4.7K
R32
47K
Q3
MMBT3904M
D19
C40
100nF
NRESET
GSM_RESET
R33
4.7K
R34
47K
Q4
MMBT3904M
D21
D22
RF_ANT
GSM_UART
GSM_TX
GSM_RX
GSM_USB
GSM_USB_D_N
GSM_USB_D_P
PWRKEY_EN
GSM_RESET
USB_VBUS
GSM_PWR
VDD_EXT
GND
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ATMEGA2560
U12
ATMEGA2560-16AU
GSM_PWR_EN
PWR12V_EN
SDMMC_DET
SD_CS
PWRKEY_EN
GSM_RESET
DRDY1
DRDY2
DRDY3
DRDY4
CS1
CS2
CS3
CS4
5V
3.3V
GND
Serial_1
MCU_RX
MCU_TX
SPI_5V
SCK
MISO
MOSI
I2C_5V
SCL_5V
SDA_5V
I2C_3.3V
SCL_3.3V
SDA_3.3V
PRGM_Serial
PRGM_RX
PRGM_TX
PRGM_DTR
Analog Inputs
J11
ADC1
ADC0
HA0200500000G
XTAL1
XTAL2
R35
1M
X1
C42
C43
20pF
X322516MLB4SI
R38
10K
RESET
SW1
PRGM_DTR
C44
100nF
J12
JTAG_TCK
JTAG_TDO
JTAG_TMS
JTAG_TDI
BH254V-10P (JTAG 2.54mm)
C45
C46
C47
C48
AREF
J13
ICSP
SB1
PB7
R41
470R
LED6
Yellow
ADC1
ADC0
D23
D24
R36
R37
4.7K
SCL_3.3V
SDA_3.3V
R39
R40
10K
SDA_5V
SCL_5V
LED5
1.5inch RGB OLED
LED7
OLED Module
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USB-to-UART for Programming





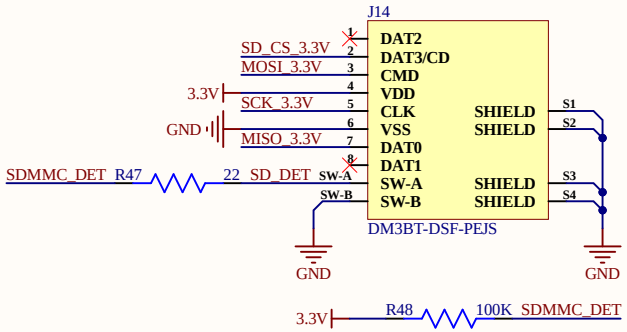


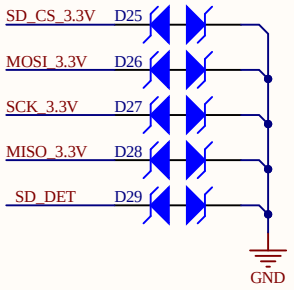


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SD Card Interface











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